

torch.ones#

<https://pytorch.org/docs/stable/generated/torch.ones.html>

Description:

Returns a tensor filled with the scalar value 1, with the shape defined by the variable argument size. size (int...) – a sequence of integers defining the shape of the output tensor. Can be a variable number of arguments or a collection like a list or tuple. out (Tensor, optional) – the output tensor. dtype (torch.dtype, optional) – the desired data type of returned tensor. Default: if None, uses a global default (see torch.set_default_dtype()). layout (torch.layout, optional) – the desired layout of returned Tensor. Default: torch.strided.

Function Signature

```
torch.ones(*size, *, out=None, dtype=None,
layout=torch.strided, device=None, requires_grad=False) →
Tensor#
```

Parameters

Keyword Arguments

Parameters

Keyword

out (Tensor, optional) – the output tensor. dtype (torch.dtype, optional) – the desired data type of returned tensor. Default: if None, uses a global default (see torch.set_default_dtype()). layout (torch.layout, optional) – the desired layout of returned Tensor. Default: torch.strided. device (torch.device, optional) – the desired device of returned tensor. Default: if None, uses the current device for the default tensor type (see torch.set_default_device()). device will be the CPU for CPU tensor types and the current CUDA device for CUDA tensor types. requires_grad (bool, optional) – If autograd should record operations on the returned tensor. Default: False.

Code Examples

```
>>> torch.ones(2, 3)
tensor([[ 1.,  1.,  1.],
        [ 1.,  1.,  1.]])

>>> torch.ones(5)
tensor([ 1.,  1.,  1.,  1.,  1.]])
```

Detailed Documentation

Documentation

Returns a tensor filled with the scalar value 1, with the shape defined by the variable argument size.

size (int...) – a sequence of integers defining the shape of the output tensor. Can be a variable number of arguments or a collection like a list or tuple.

out (Tensor, optional) – the output tensor.

dtype (torch.dtype, optional) – the desired data type of returned tensor. Default: if None, uses a global default (see `torch.set_default_dtype()`).

layout (torch.layout, optional) – the desired layout of returned Tensor. Default: `torch.strided`.

device (torch.device, optional) – the desired device of returned tensor. Default: if None, uses the current device for the default tensor type (see `torch.set_default_device()`). device will be the CPU for CPU tensor types and the current CUDA device for CUDA tensor types.

requires_grad (bool, optional) – If autograd should record operations on the returned tensor. Default: False.